

	Work Instruction	Doc. Revision	01
	AXI V810 S2EX X-Ray Tube Alignment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	29 Apr 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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X-Ray Tube Alignment Steps:

1. Unload the board from machine is there is any.
2. Go to X-Ray Source tab and turn off the X-Ray as shown.

Services → Subsystem Status & Control → X-Ray Source → Turn off Current and Voltage

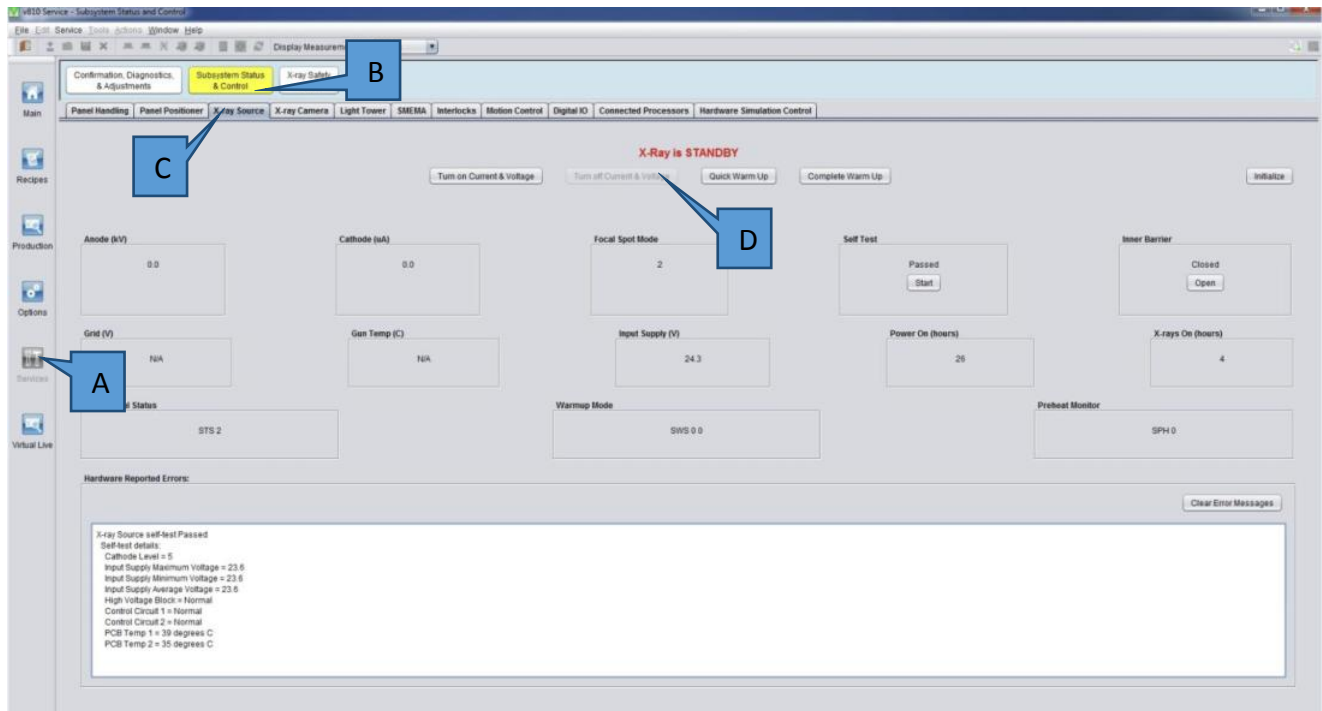


Figure 1

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3. Go to Panel Handling tab, click Initialize and ensure the AWA is completely home (19 inches).

Services → Subsystem Status & Control → Panel Handling → Initialize → Home Rails → Open Inner Barrier

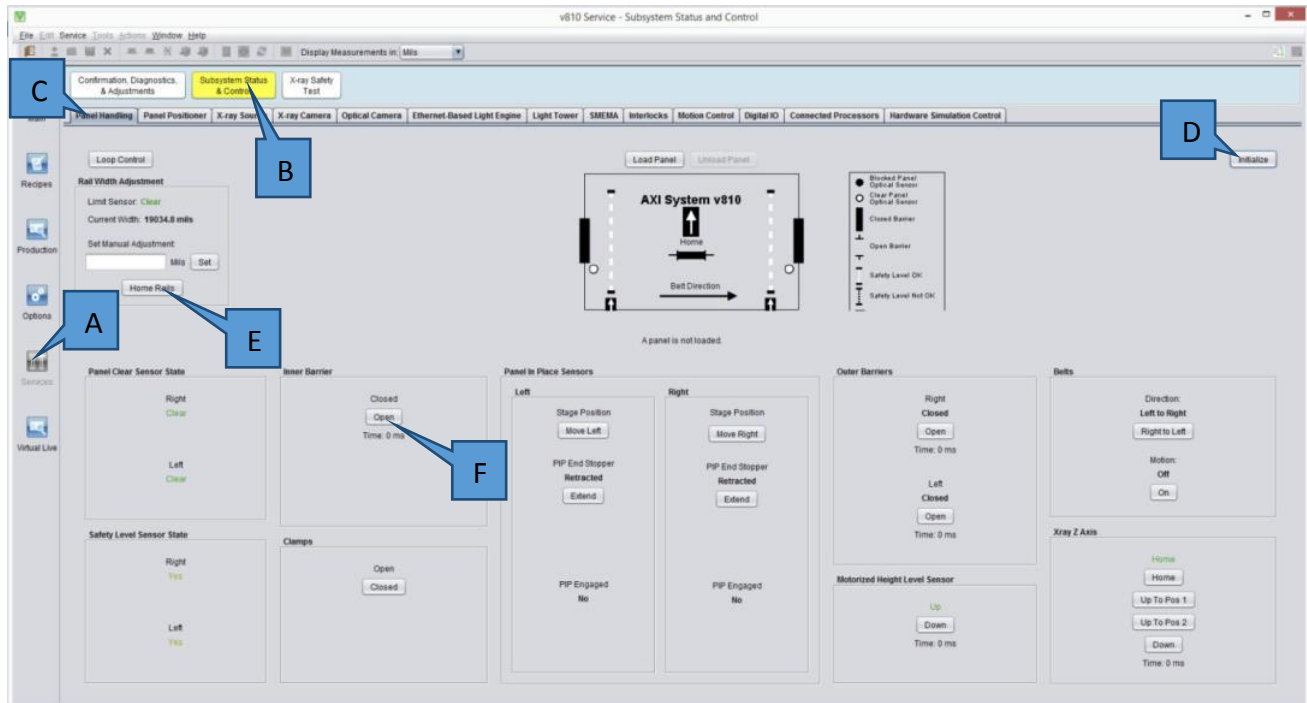


Figure 2

4. Turn off the X-Ray key at Operator Console and quit the close the V810 GUI. Open front door and move the XY stage to the middle. Unscrew the 4 point and remove the FIELD STOP ASSEMBLY XXL (P/N, 4LAGR-A045) top cover as shown.



Figure 3

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5. Prepare the X-Ray Alignment jig as shown.

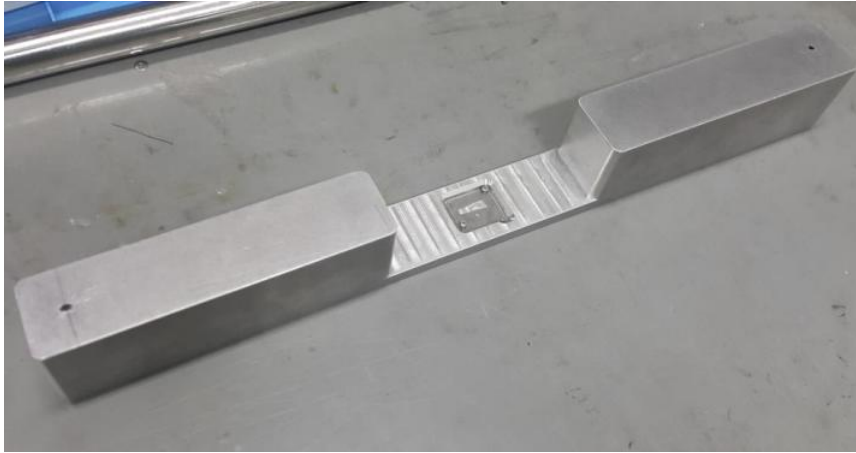


Figure 4

6. Place the jig on top of the Field Stop Assembly as shown. Remark on the dowel pin orientation and make sure the jig is place flat and seat firmly.

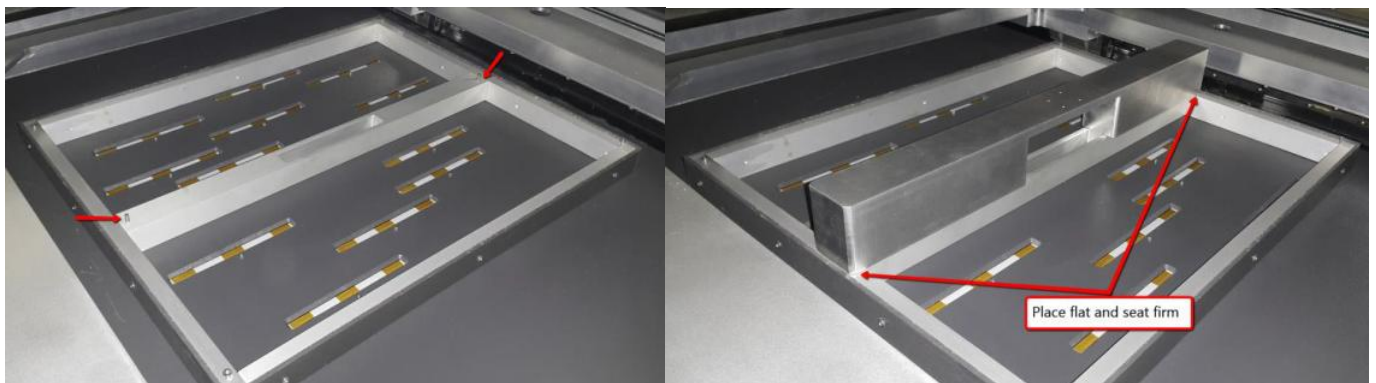


Figure 5

7. Go to the top root and open the top access door as shown Figure 6.



Figure 6

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8. Check the X and Y axis locking screws and ensure the screw is located centre of the track. If not, loosen the screw and adjust the tube XY base plate with turning the ADJUSTMENT KNOB ASSEMBLY (P/N, 4XAHW-A008).

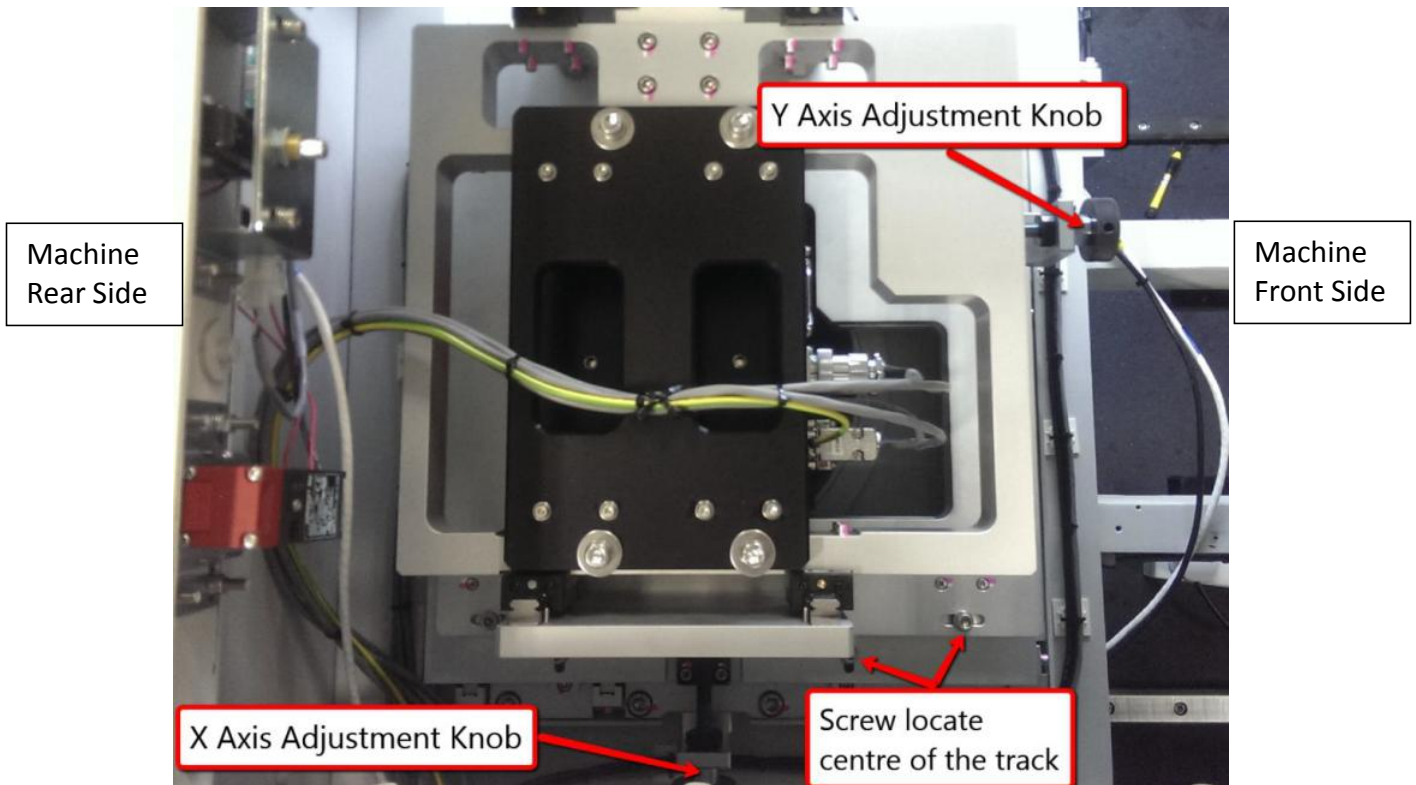


Figure 7

9. Once done, close all entire access doors and launch the MFX application which located at desktop.



Figure 8

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10. Ensure the COM 2 is selected (show at the bottom left corner), else, go to Tools → Option → Comm → Select Com 2 → OK. Once done, go to Tools and click connect.

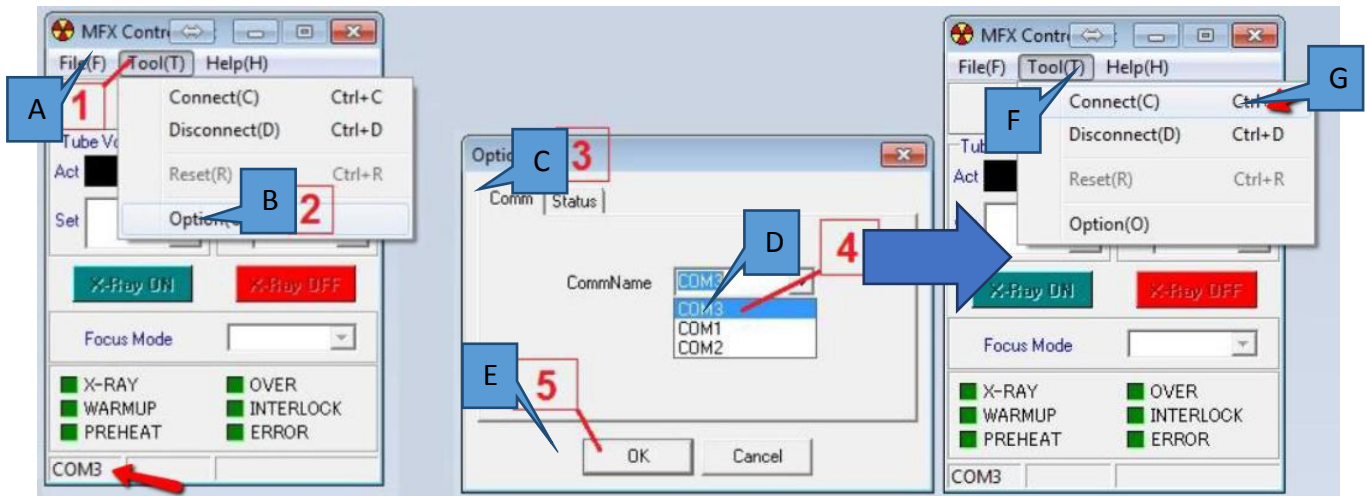


Figure 9

11. Select the “Large” focus mode and ensure the setting was X-ray tube Voltage (130kv) and Current (200uA) as shown in the Figure 16. Press X-ray ON and leave it for warm-up for 1 hour

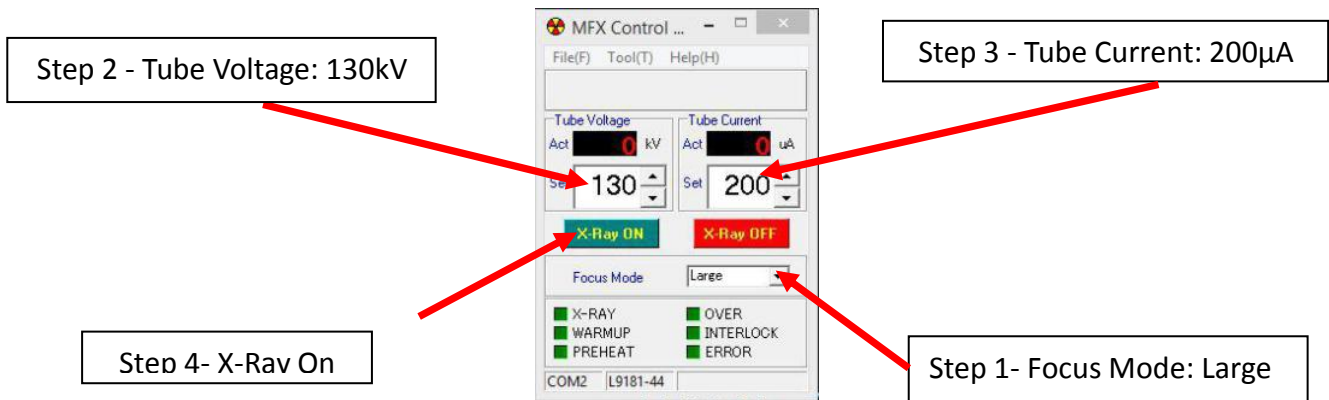


Figure 10

12. Download and unzip the “BehnkeTool” application and paste in C Drive. Ensure the “tftpboot” is the master folder.

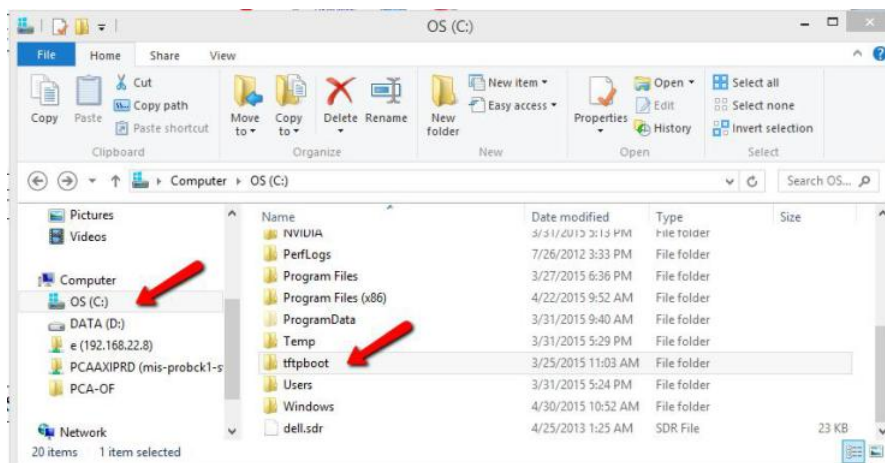


Figure 11

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13. Go to path “C:\tftpboot\BehnkeTool v2” and launch the “area mode Camera 0” to continue X-ray tube positioning.

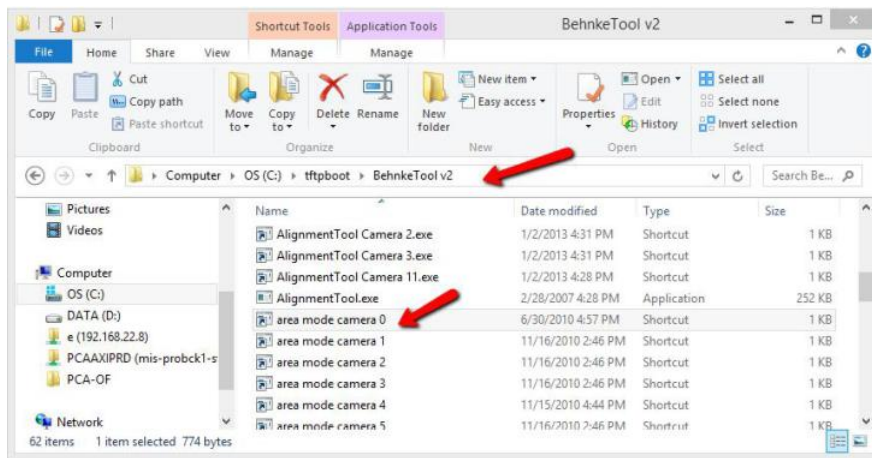


Figure 12

14. Once launch the application, will see screen similar to below.

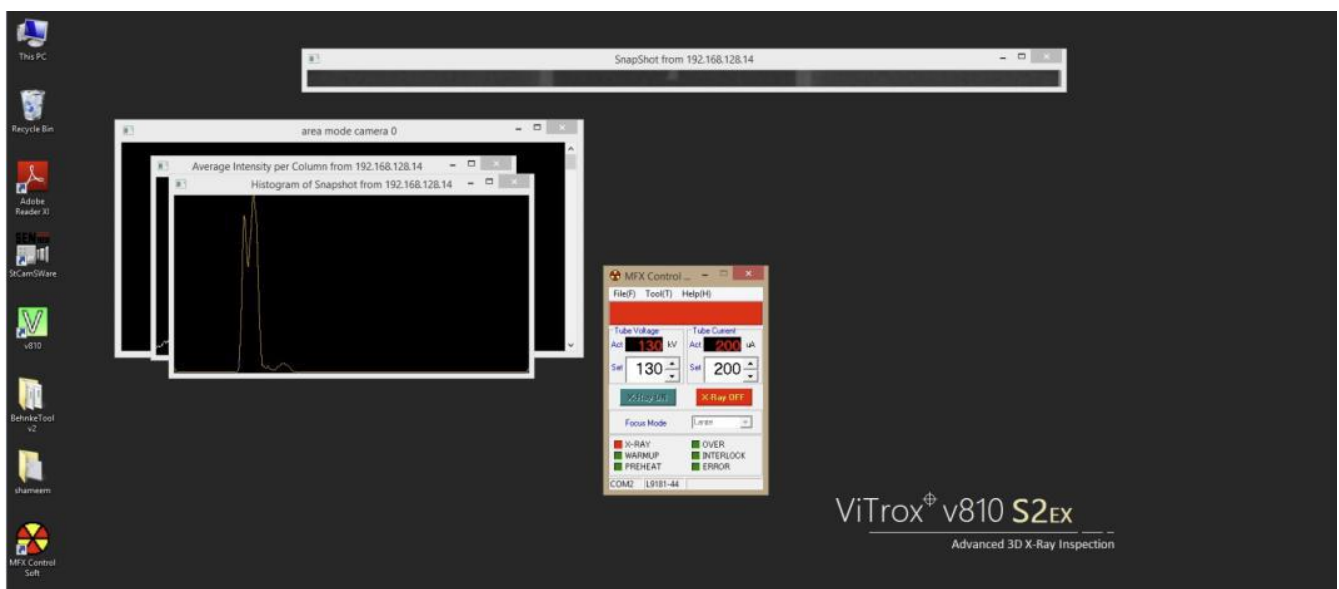


Figure 13

15. Open a “Wordpad” blank document and zoom into 190% ratio as shown.

Ease for tube positioning

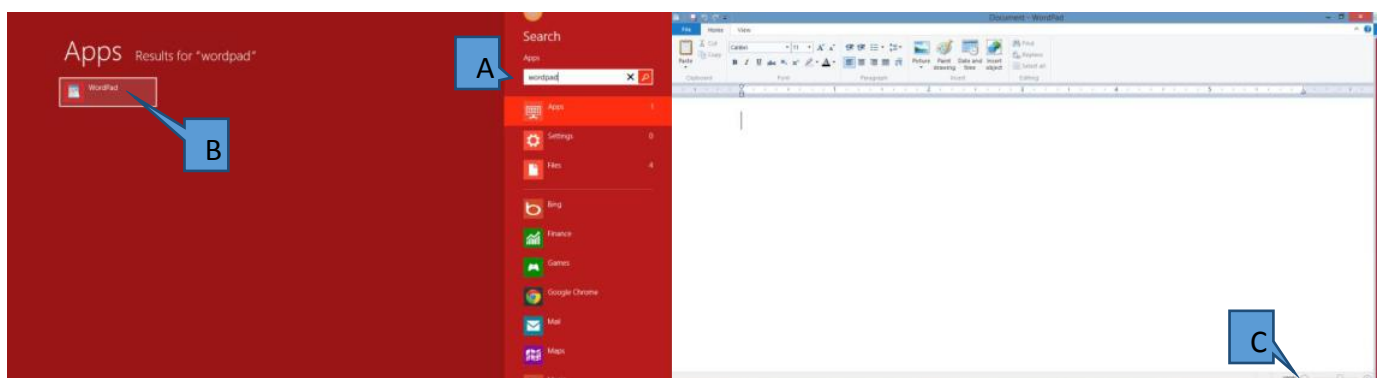


Figure 14

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16. Drag the Camera 0 area mode live window on top of the “word pad” as shown. Ensure it place precision at the ruler bar.

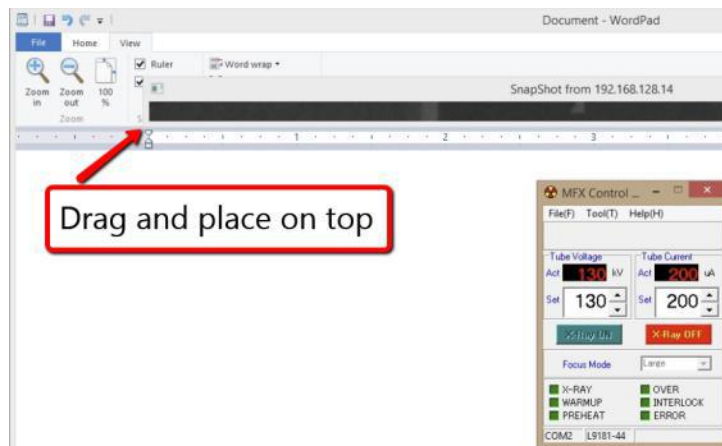


Figure 15

17. Check on the area mode image and optimize the **X-Axis** until the triangle edge located at the centre on image.

Sample; Image below need to move physical the tube X axis toward to machine left side as need to get the triangle edge to centre of image.

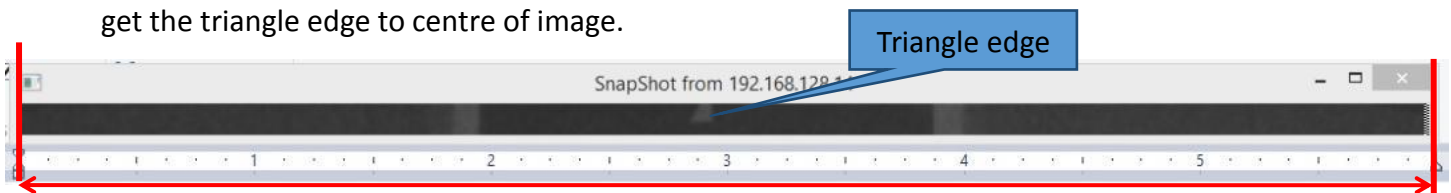


Figure 16

18. Turn off the X-ray source at MFX tools and key at control panel, open top access door. Loosen the screws and adjust the X Axis Adjustment knob. # 1 turn of knob will be 40 mils (1 mm).

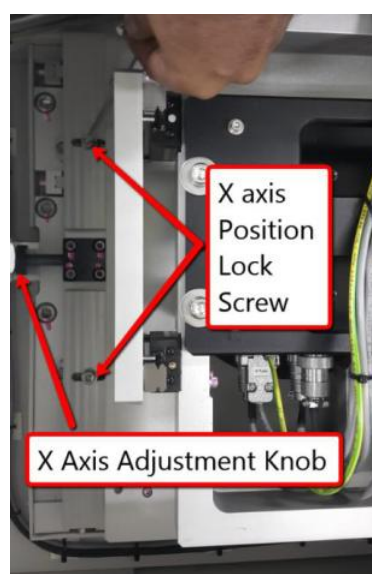


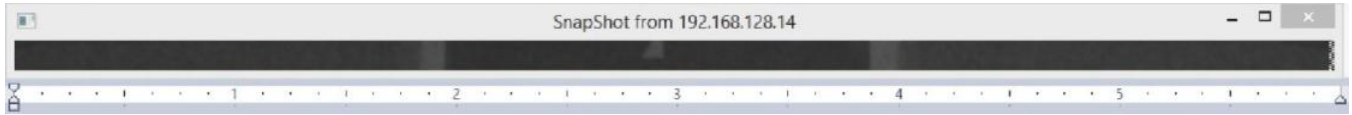
Figure 17

Term, If wish to *move image to right, tube need to move physically to the left* or *move image to left, tube need to move physically to the right.*

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19. Once done, tighten the lock screws and close the access panel. Turn ON the key and power up tube at MFX application, check on the latest image for area mode cam 0. Repeat step 17 and 18 until the spot is optimizing. Below image showing the before and after optimized.

Before Adjustment



After Adjustment

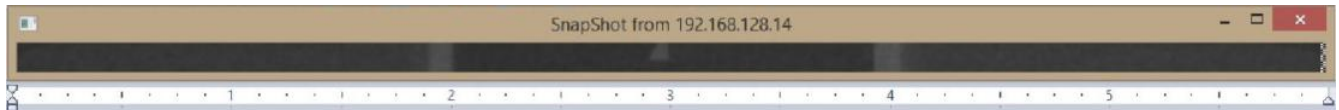


Figure 18

20. Check on the area mode image and optimize the **Y-Axis** until the triangle edge located at the centre on image.

Sample; Image below need to move physical the tube Y axis toward to machine front side as need to get the triangle edge to centre of image.

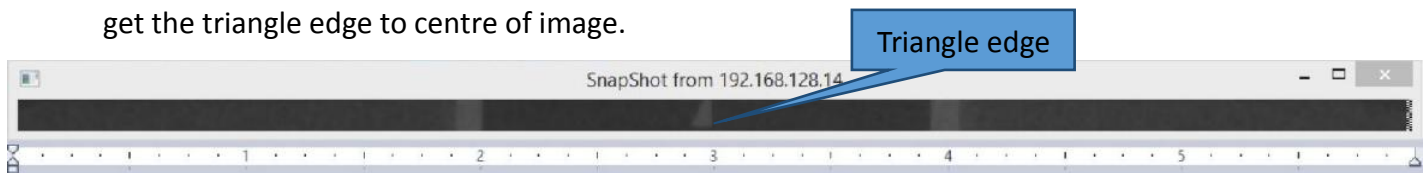


Figure 19

21. Turn off the X-ray source at MFX tools and key at control panel, open top access door. Loosen the screws and adjust the Y Axis Adjustment knob. # 1 turn of knob will be 40 mils (1 mm).

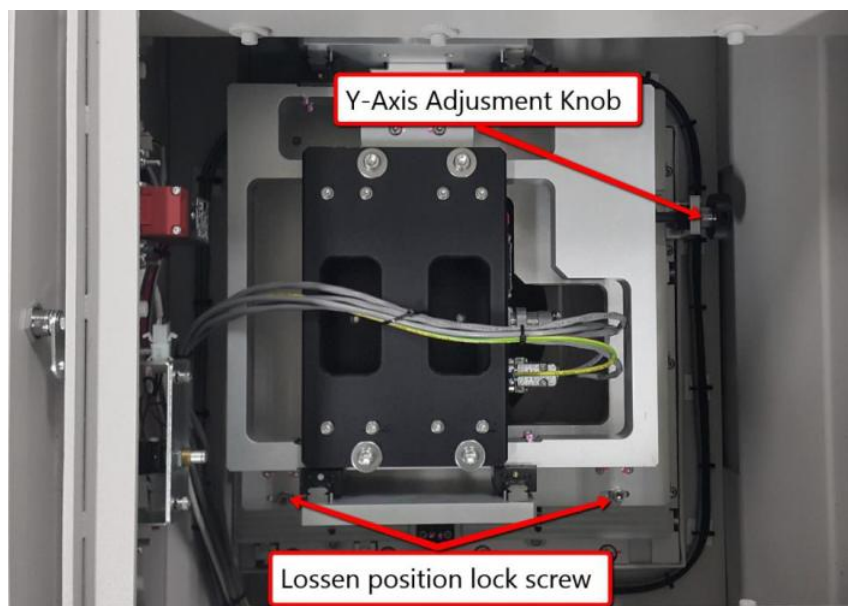


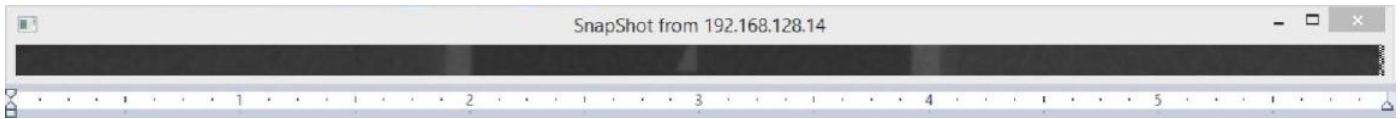
Figure 20

Term, If wish to *move image to top*, tube need to move physically to the front or move image to *bottom*, tube need to move physically to the rear.

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22. Once done, tighten the lock screws and close the access panel. Turn ON the key and power up tube at MFX application, check on the latest image for area mode cam 0. Repeat step 20 and 21 until the spot is optimizing. Below image showing the before and after optimized.

Before Adjustment



After Adjustment

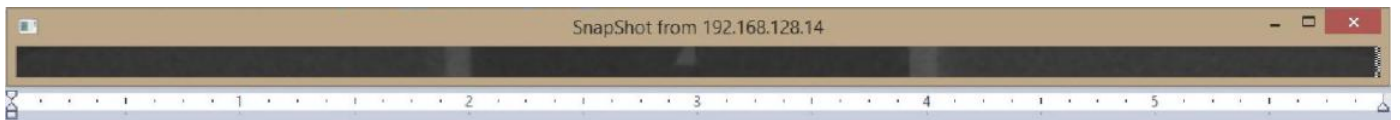


Figure 21

23. Once done optimize X and Y axis, click stop and go to Tools → Disconnect. Turn off the key at control panel as well.

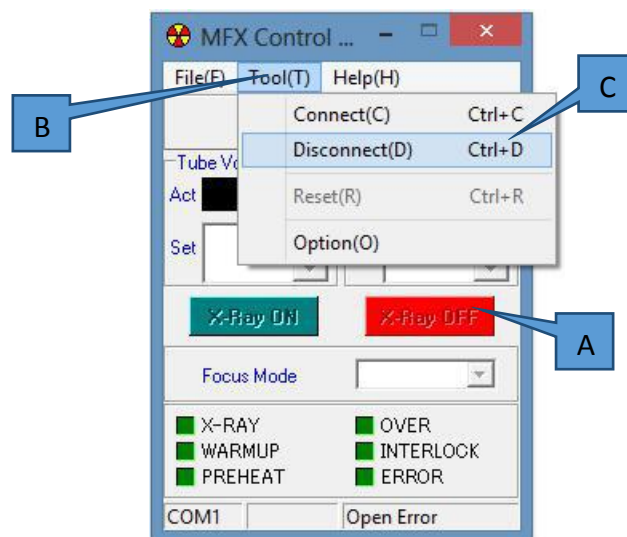


Figure 22

24. Open the front access door and remove the jig from machine and assembly the FSA cover back to origin position.



Figure 23

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25. Go to path “C:\Program Files\AXI System\v810\5.0\s2exConfig” and open the “**software.config**”.
- Amend the key “**saveAllConfirmationImages**” to true and save the config file.

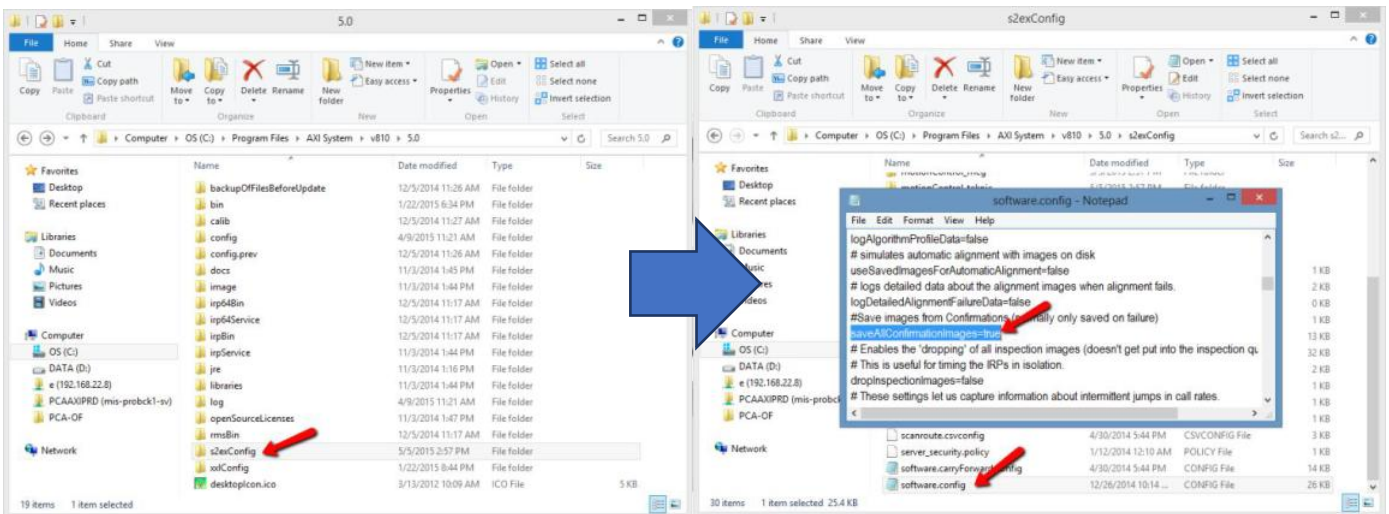


Figure 24

26. Launch the V810 GUI, at the main tab and ensure the 19um is selected as low mag setting. If not selected, manually change the setting. Need to close and restart GUI once setting change.
- Mag change icon → Low Mag setting(19um)→ ok → Restart GUI*

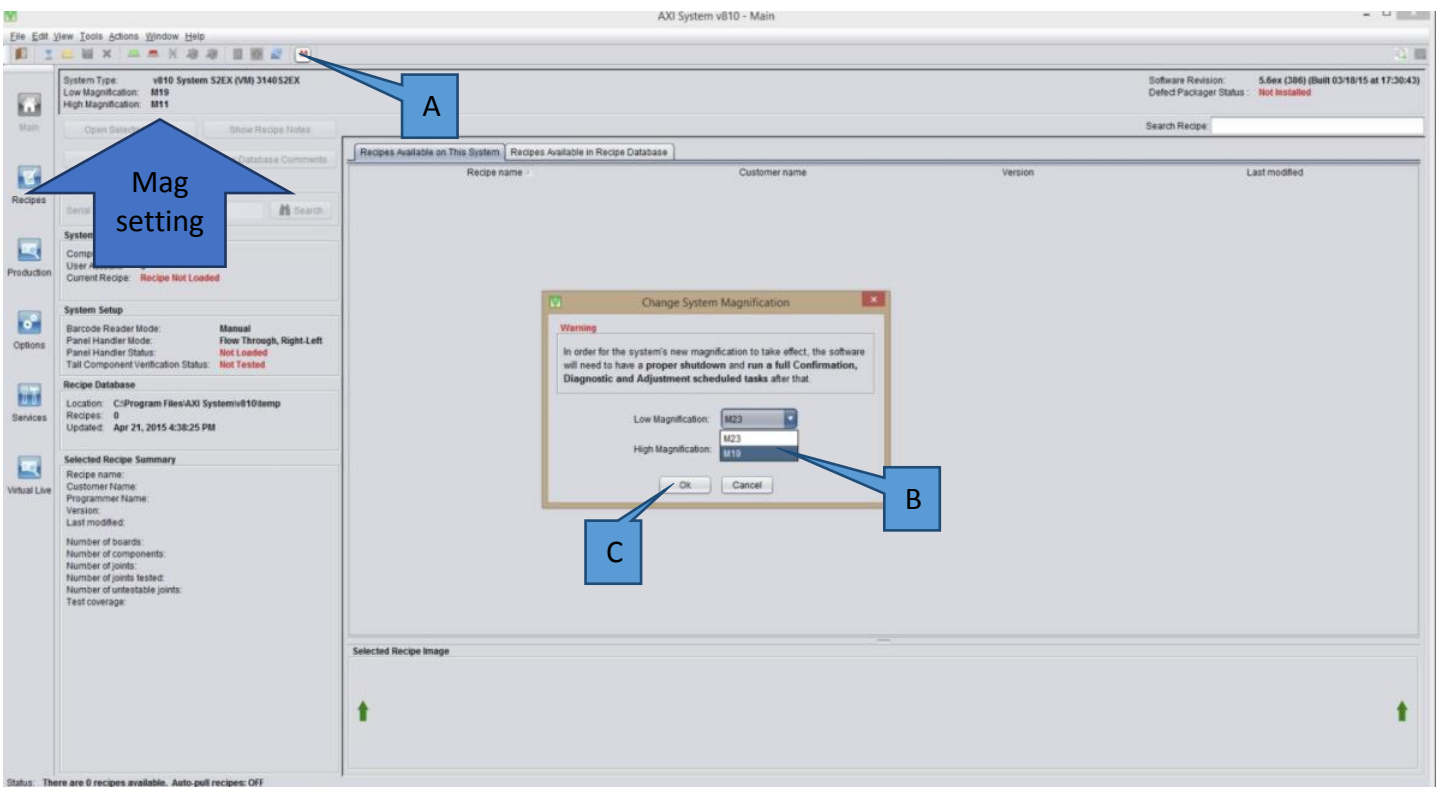


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27. Go to Confirmation, Diagnostic and Adjustment tab, select the “Schedule Task” and run for 5 loops and ensure no any failure occurrence.

Services → CDnA → Schedule Task → 5 loops → Run

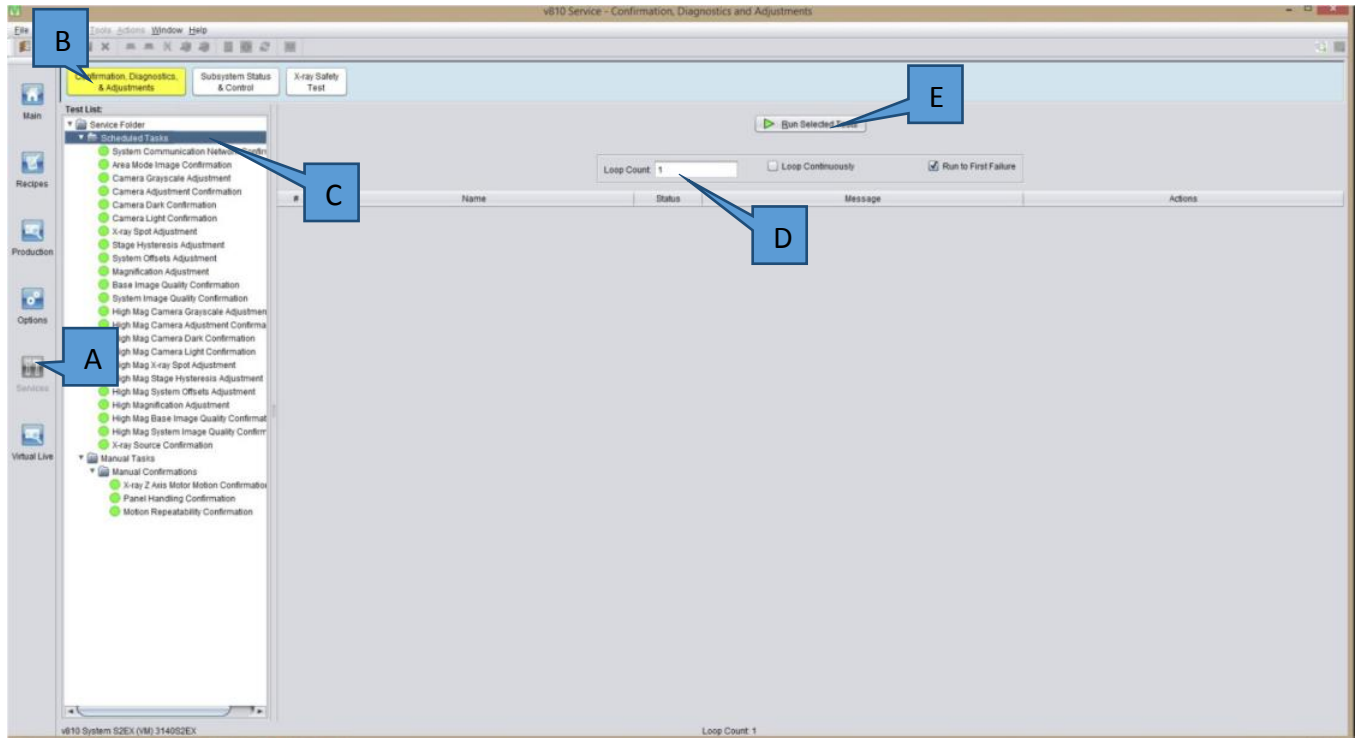


Figure 26

28. Once done, go to path “C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot” and check for the X-Ray spot image. Should see the X-ray spot similar as below (**Reference only**).

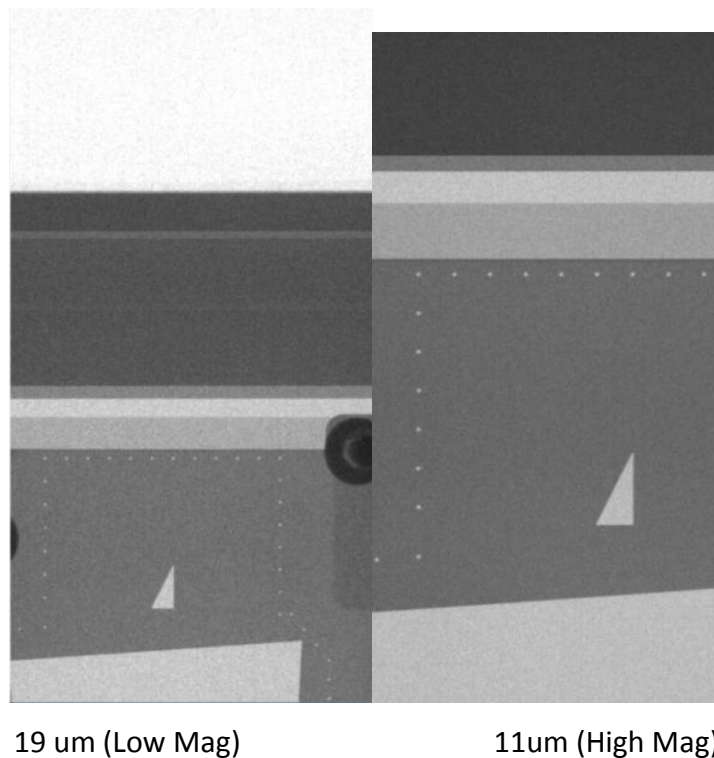


Figure 27

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29. Back to the main tab, select the 23um as low mag. Once change, close and restart GUI once setting change.

Mag change icon → Low Mag setting (23um) → ok → Restart GUI

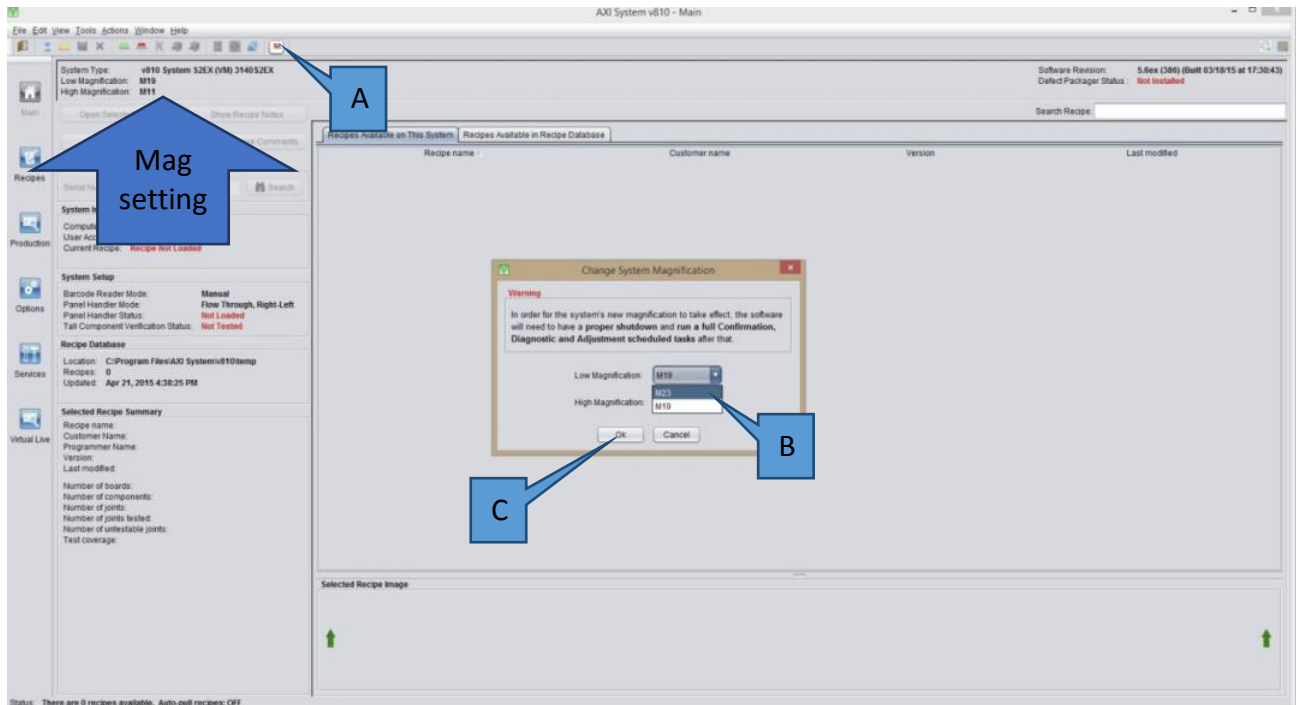


Figure 28

30. Go to Confirmation, Diagnostic and Adjustment tab, select the “Schedule Task” and run for 5 loops and ensure no any failure occurrence.

Services → CDnA → Schedule Task → 5 loops → Run

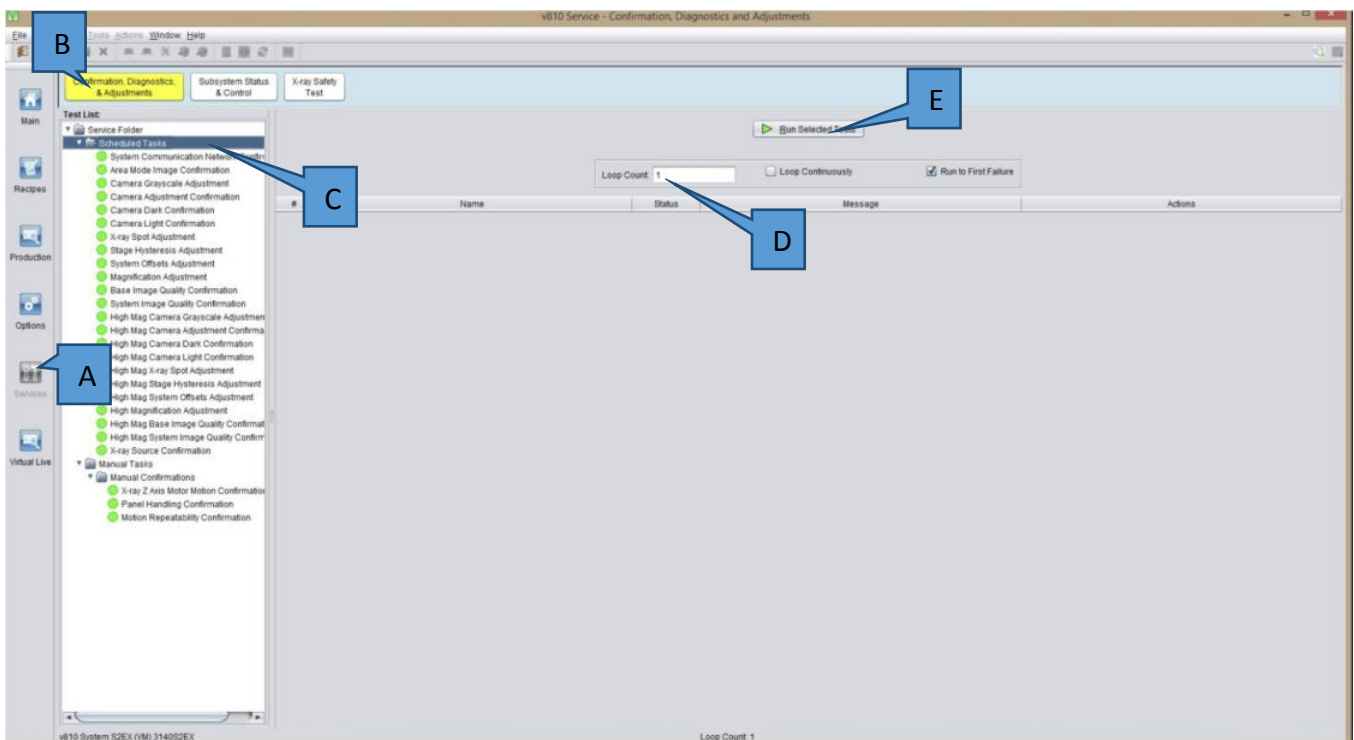
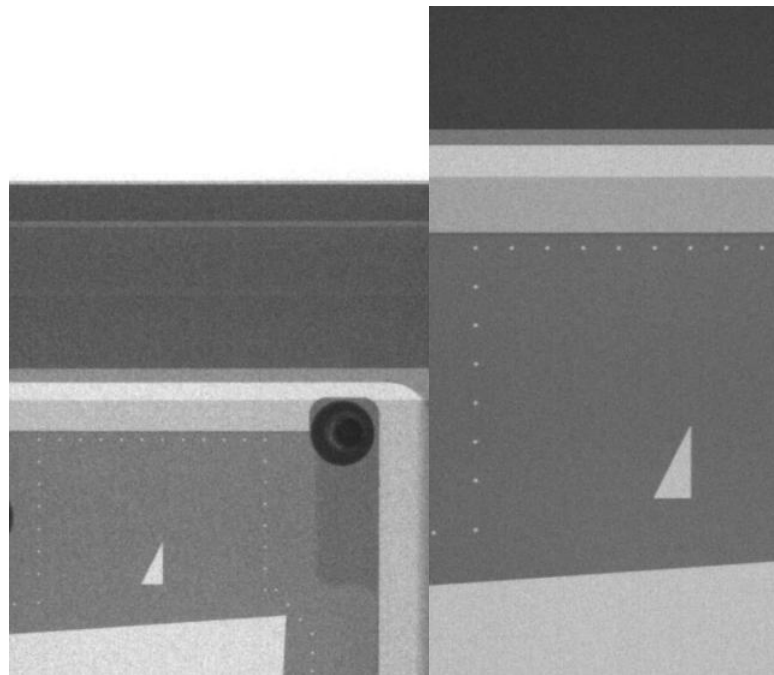


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31. Once done, go to path “C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot” and check for the X-Ray spot image. Should see the X-ray spot similar as below **(Reference only)**.



23 um (Low Mag)

11um (High Mag)

Figure 30